

Feiyang Huang

PhD Candidate, Computational Biology & Medicine

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 🌐 fyng 📄 feiyang-huang 🗣 English, Mandarin

Education

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| PhD Weill Cornell Graduate School of Medical Sciences , Tri-Institutional PhD Program in Computational Biology & Medicine
• Advisor: Wesley Tansey, Memorial Sloan Kettering Cancer Center | New York, NY
Aug 2023 – present |
| BS Johns Hopkins University , Computer Science and Biomedical Engineering
• GPA 3.96
• Activities: President, Singapore Students' Association (2021–2023); Theta Tau Professional Engineering Fraternity (2020–2022) | Baltimore, MD
Sept 2020 – May 2023 |

Experience

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| Memorial Sloan Kettering Cancer Center, Tansey Lab , PhD Researcher
• Developing OncoState, a self-supervised transformer learning patient states from multimodal clinical trajectories of 99,000+ MSK-IMPACT patients; outperforms the IMDC risk model in advanced clear cell RCC.
• Co-led a population-scale study of clinical and genetic determinants of systemic-therapy toxicity (co-first author, medRxiv 2026): developed the random forest toxicity model and the analysis and narrative for Figure 3.
• Contributed to PPC , a pan-cancer atlas of ex vivo drug screens for functional precision oncology. | New York, NY
June 2024 – present |
| Memorial Sloan Kettering Cancer Center , PhD Rotations
• Tansey Lab : Bayesian active learning for nominating combination therapies from ex vivo drug screens
• Chodera Lab : Equivariant graph neural networks for small-molecule property prediction (mtenn)
• CMO Innovation Lab : Computational pipeline for bisulfite- and enzyme-based DNA methylation sequencing of tumors
• Morris Lab : BERT representation learning from tumor mutation profiles | New York, NY
Aug 2023 – May 2024 |
| Johns Hopkins University, Fan Lab , Undergraduate Researcher
• Built and benchmarked a reference-free LDA pipeline to deconvolve spatial transcriptomics data (STdeconvolve, <i>Nature Communications</i> 2022).
• Developed a torque-based method and simulation framework to quantify subcellular mRNA polarization. | Baltimore, MD
Jan 2021 – May 2023 |
| Hologic , Software Engineering Intern
• Built an internal work-history search tool and automated release-note generation (3 h to 20 min). | May 2022 – Aug 2022 |
| Johns Hopkins University, Biomedical Design , Team Lead, iMEDS
• Led a team of 7 building an algorithm to continuously monitor pediatric ICU sedation; interviewed 20+ clinicians. | Baltimore, MD
Aug 2022 – May 2023 |
| Johns Hopkins University, Institute for NanoBioTechnology , Research Intern
• Optimized a point-of-care PCR assay (15% lower detection limit) and its embedded image-analysis and motor-control code. | Baltimore, MD
May 2021 – Aug 2021 |
| DioTeX Diagnostics , Co-Founder
• Led assay development for a lateral-flow test for internal hemorrhage; raised over \$60,000 in non-dilutive funding. | Baltimore, MD
Nov 2020 – Aug 2023 |
| National University of Singapore, Yu Lab , Research Intern
• Trained a CNN for endoscopic image classification (91% accuracy); co-author in <i>Gastrointestinal Endoscopy</i> . | Singapore
Nov 2019 – Dec 2020 |
| Singapore Armed Forces , 3rd Sergeant
• Led a detachment of 3 soldiers and co-supervised daily training for 15. | Singapore
Jan 2018 – Oct 2019 |

Publications

- Population-Scale Precision Safety in Oncology Reveals Clinical and Genetic Determinants of Systemic Therapy Toxicity** Sept 2026
Ziad Bakouny*, X Alex Guo*, **Feiyang Huang***, Saksham Mohan*, Rohan Walser*, et al., Wesley Tansey, Jian Carrot-Zhang, Ed Reznik
www.medrxiv.org/content/10.64898/2026.09.16.26363259 (medRxiv preprint; *equal contribution)
- A Pan-Cancer Ex Vivo Drug Screen Atlas for Functional Precision Oncology** Feb 2026
Karl Pichotta, Jessica B White, Jeffrey F Quinn, Anneliese Markus, Christopher Tosh, Antoine De Mathelin, Erin Coyne, **Feiyang Huang**, Wesley Tansey
pmc.ncbi.nlm.nih.gov/articles/PMC12934811 (bioRxiv preprint)
- Reference-free cell type deconvolution of multi-cellular pixel-resolution spatially resolved transcriptomics data** Apr 2022
Brendan F Miller, **Feiyang Huang**, Lyla Atta, Arpan Sahoo, Jean Fan
[10.1038/s41467-022-30033-z](https://doi.org/10.1038/s41467-022-30033-z) (Nature Communications 13, 2339)
- Artificial intelligence-enhanced white-light colonoscopy with attention guidance predicts colorectal cancer invasion depth** Sept 2021
Xiaobei Luo, Jiahao Wang, Zelong Han, Yang Yu, Zhenyu Chen, **Feiyang Huang**, et al., Hanry Yu
[10.1016/j.gie.2021.03.936](https://doi.org/10.1016/j.gie.2021.03.936) (Gastrointestinal Endoscopy 94(3), 627–638)
- Roles and regulation of long noncoding RNAs in hepatocellular carcinoma** Oct 2019
Lee Jin Lim, Samuel YS Wong, **Feiyang Huang**, et al., Caroline G Lee
[10.1158/0008-5472.CAN-19-0255](https://doi.org/10.1158/0008-5472.CAN-19-0255) (Cancer Research 79(20), 5131–5139)

Conference Abstracts

- Bakouny Z, Guo XA, Walser R, **Huang F**, et al. Pan-cancer discovery of clinical and genomic determinants of immune-related adverse events. *Journal of Clinical Oncology* 44(16_suppl):11132 (ASCO 2026).
- Wu S, Pejic B, Roy S, **Huang F**, et al. Development of a computational model of SBS scores in critically ill children. *Pediatric Critical Care Medicine* 25(11S):e87 (2024).

Patents

- Parent C, Simon E, **Huang F**, Ruci A, Zhang E. Methods and devices for quantitatively estimating thrombomodulin. US Patent App. 19/240,058 (2025).
- Eng R, Parent C, Hayat A, Ruci A, Krishnan A, **Huang F**, Simon E, Zhang E. Methods and devices for quantitatively estimating syndecan-1. US Patent App. 17/715,294 (2022).

Presentations

- 45th Vincent du Vigneaud Research Symposium, Weill Cornell Medicine, New York, NY (2026). **Huang F**, Tansey W. OncoState: learning patient states from clinical trajectories in cancer. Poster #66.
- Military Health System Research Symposium, Kissimmee, FL (2022). Towards point-of-care internal hemorrhage detection through a syndecan-1 lateral flow assay. Oral presentation.
- Johns Hopkins BME Design Day, Baltimore, MD (2022). DioTeX: Hemorrhage Diagnostics. Poster.

Honors and Awards

- NIH DEBUT Challenge, Healthcare Technologies for Low-Resource Settings Prize** 2023
With the DioTeX team; \$15,000.
- VentureWell e-Fest Prize** 2023
With the DioTeX team; \$10,000.

Teaching

- Course Assistant, EN.601.230 Mathematical Foundations for Computer Science, Johns Hopkins University (Fall 2022).